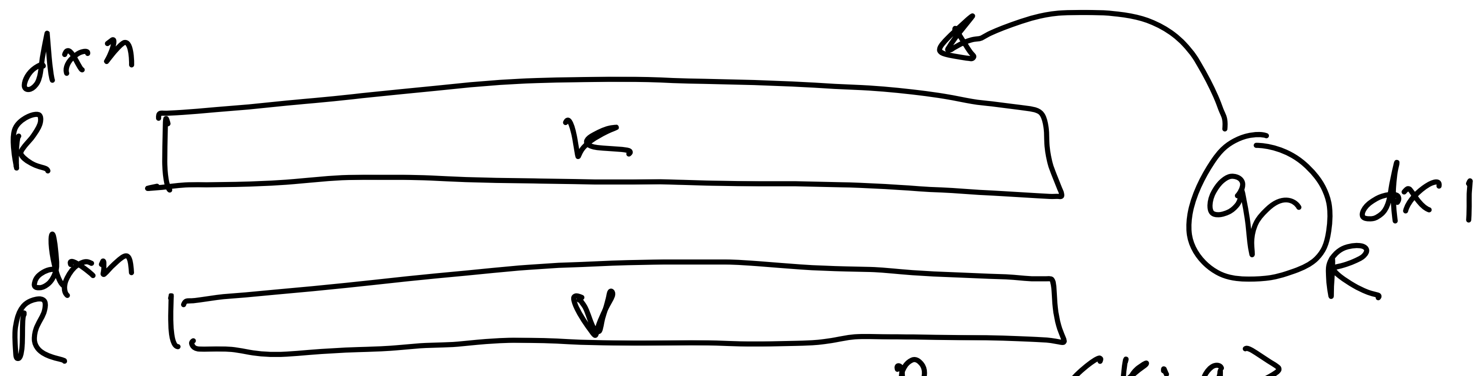
Example of where to use the Rejection Sampling (Cooked up)

Consider Attention Computation:



$$\mathrm{Att}(K, V, q) = \frac{\sum_{1}^{n} e^{\langle k_i, q\rangle} v_i}{\sum_{1}^{n} e^{\langle k_i, q\rangle}}$$

Let us say you wanted to approximate the denominator with a sample

$$S = \sum_{1}^{n} e^{\langle k_i, q\rangle}$$

What is the best sampling probability distribution?

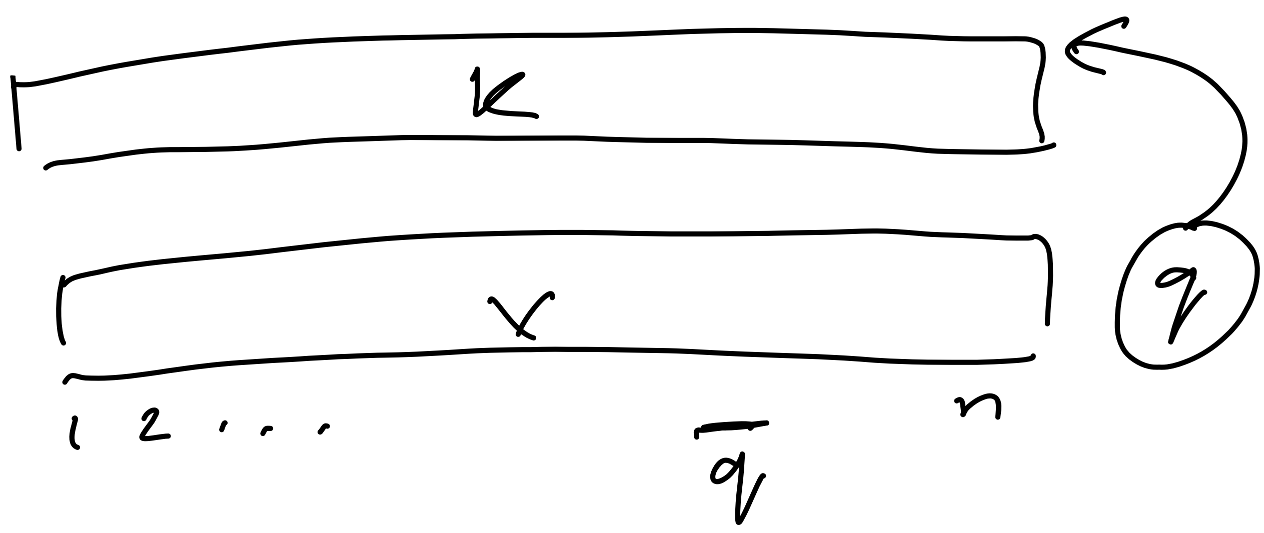
→ Importance Sampling say

$$p(i) \propto e^{\langle k_i, q \rangle}$$

Say some how we have obtained the sample proportional to

$$\langle k_i, \bar{q} \rangle$$

where $\bar{q}$ is some "representative" query. Note q is not available before hand in decode.



$$[i_1, i_2 \ldots i_n]$$

$$p(i_j) \propto \langle k_{i_j}, \bar{q} \rangle$$

Now can you get a sample from this subset that is s.t

$$p(i_{j_k}) \propto \langle k_{i_{j_k}}, q \rangle$$

where q is the actual query?

Can use Rejection Sampling!!